

INMO(17th) – 2002

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. For a convex hexagon ABCDEF, consider the following six statements: (i) AB is parallel to DE (ii) BC is parallel to EF (iii) CD is parallel to FA (iv) AE = BD (v) BF = CE (vi) CA = DF.
 - (a) Show that if all the six statements are true, then the hexagon is cyclic.
 - (b) Prove that, in fact any five of these six statements also imply that hexagon is cyclic.
 2. Determine the least positive value taken by the expression $a^3 + b^3 + c^3 = 3abc$ as a, b, c vary over all positive integers. Find also all triples (a, b, c) for which this least value is attained.
 3. Let x, y be positive real such that $x + y = 2$. Prove that $x^3 y^3 (x^3 + y^3) \leq 2$.
 4. Do there exist 100 lines in a plane, no three of them are concurrent. Such that they intersect exactly in 2002 points?
 5. Do there exist three distinct positive real numbers a, b, c such that the numbers $a, b, c, b+c-a, c+a-b, a+b-c, a+b+c$ form a 7 term arithmetic progression in some order?
 6. Suppose the n^2 numbers $1, 2, 3, \dots, n^2$ are arranged to form an n by n array consisting of n rows and n columns such that the numbers in each row (from left to right) and each column (from top to bottom) are in increasing order. Denote by a_{jk} the number in j th row and k th column. Suppose b_j is the maximum number of entries that can occur as a_{jj} , $1 \leq j \leq n$. Prove that $b_1 + b_2 + \dots + b_n \leq \frac{n}{3}(n^2 - 3n + 5)$. (Example: in case $n = 3$, the only number which can occur as a_{22} are 4, 5 or 6 so that $b_2 = 3$.)
